

John Okyere

Full-Stack Engineer | AI & Web3

✉ johnokyere282@icloud.com ☎ +233591518686 📍 Remote | GMT (UTC) 🔗 johnokyere.xyz
🐙 github.com/mhiskall282 🔗 linkedin.com/in/johnokyere

PROFESSIONAL SUMMARY

Full-Stack Engineer with 3+ years of experience building end-to-end AI-native Web3 products. Specialized in TypeScript, React/Next.js, Node.js, PostgreSQL, AI agents & LLM integrations, and Solidity smart contracts. Driven by how products work and feel bridging engineering, product, and UX. Proven track record training 500+ developers across blockchain ecosystems including Ethereum, ICP, and Hyperledger Fabric. Winner of 4 international hackathons.

TECHNICAL SKILLS

Languages TypeScript, JavaScript, Solidity	Frontend - React, Vercel, Tailwind CSS, Shadcn UI, Radix Ui, - Framer Motion. Zustand, TanStack Query
Backend & APIs Node.js, Bun, Express.js, Hono, ElysiaJS, REST APIs, WebSockets, Zod, Redis / Upstash, FastAPI, Rust	Databases & Storage PostgreSQL, Supabase, Neon, Drizzle ORM, Prisma, pgvector, Pinecone, IPFS (Pinata), AWS RDS
AI & Agents OpenAI, Claude, Gemini, OpenRouter, Vercel AI SDK, LangChain, AI agents, RAG, MCP	Blockchain & Web3 Ethereum (EVM), Solana (SVM), Foundry, Hardhat, viem, wagmi, Privy, ICP, Sui
Tooling & Infrastructure Git, GitHub Actions (CI/CD), Vercel, Docker, Turborepo, AWS, GCP, Azure, EC2, S3, Render, Railway	Product & Design Figma, design systems, product thinking, user testing

PROFESSIONAL EXPERIENCE

Technical Lead, Kayaba Labs  - Built and shipped a full-stack LMS using Next.js, Node.js, and PostgreSQL from concept to production, scaling to 500+ active learners within 4 months of launch. - Designed and delivered technical training programs for 30+ developers across web development, AI, and blockchain, resulting in 85% skill certification rate and 20+ deployed projects. - Architected and delivered full-stack solutions using TypeScript, React, and Node.js, managing 5+ concurrent projects and achieving 99.9% uptime across production systems.	09/2025 – Present Accra, Ghana
Resident Fellow, Ethereum Ethiopia - Selected from 200+ global applicants for competitive Ethereum builder fellowship, delivering 3 technical milestones, including an AI-powered evidence verification system and EVM-based carbon credit smart contracts over 6 weeks. - Led full-stack development for EcoFusion carbon credit platform, implementing AI verification with 90%+ accuracy and tokenizing 500+ sustainability actions on Hedera DLT during a 6-week build sprint.	01/2025 – 02/2025 Addis Ababa, Ethiopia

- Achieved top 5 ranking among 40+ fellowship projects by implementing a multi-modal AI verification pipeline on Hedera DLT that validated carbon credit claims with 90%+ accuracy recognized by judges for combining AI and on-chain trust mechanisms.

Founding Engineer, CredAxis

02/2025 – 12/2025

Remote

- Built and launched decentralized credential verification platform from concept to live beta in 3 months, serving 200+ users with sub-3-second on-chain verification and 100% tamper-proof record integrity.
- Architected full-stack platform using Next.js, Node.js, and Solidity, deploying smart contracts that processed 500+ verification requests and achieving sub-100ms frontend response times through optimized UX flows.

AOC DevConnect Scholar, DevConnect Argentina (EF)

09/2025 – 10/2025

Buenos Aires, Argentina

- Secured \$1,000 competitive scholarship (1 of 20 African recipients) to represent African builders at Devconnect Argentina, establishing 8+ partnerships with Ethereum core teams that led to ongoing collaboration on the Base Carbon Meter DePIN project.
- Collaborated with 50+ Ethereum and Celo core developers across 10+ technical workshops on ZK cryptography and DeFi infrastructure, applying learnings to architect the ZK verification layer for Base Carbon Meter and the ZK-proof components of CredAxis.
- Gained applied expertise in ZK-SNARKs and proof systems, architecting a ZK verification layer for Base Carbon Meter that processed energy telemetry with 99.8% tamper-proof validation accuracy, reducing fraudulent data risk by an estimated 40%.

ZK Scholar & Resident Fellow, Dev3Pack Global

09/2025 – 10/2025

Palermo Soho,
Argentina

- Completed competitive ZK Scholar fellowship, mastering Groth16 proof systems and ZK-SNARKs, then immediately deploying a production ZK verification layer that processed 50GB+ of live energy telemetry data with 99.8% validation accuracy for Base Carbon Meter.
- Joined Base Carbon Meter DePIN project through fellowship, contributing full backend infrastructure and ZK verification layer that enabled tokenization of 1,000+ verified renewable energy offsets on Base.
- Built backend AWS infrastructure using S3, Lambda, and DynamoDB, processing 50GB+ of energy telemetry data with 99.8% validation accuracy through a custom ZK verification layer integrated with Amazon Bedrock LLM routing.

ICP Ambassador, ICP Hubs Global

01/2025 – 12/2025

Central Region, Ghana

- Mentored 200+ students and developers on building blockchain solutions using Motoko and Rust, resulting in 40+ prototype deployments on the Internet Computer network.
- Collaborated with global ICP Hub teams to create localized educational content on decentralized apps and canister smart contracts, increasing developer participation by 25%.
- Promoted ICP technology adoption across Ghana by organizing 5+ developer workshops and community events, reaching 600+ participants and growing local engagement by 40%.

CyberSecurity Intern, Pinnacle Labs

06/2024 – 08/2024
Bengali, India

- Built password strength analyzer evaluating 10,000+ passwords with 94% vulnerability detection accuracy, implementing AES-256, DES, and RSA encryption algorithms to harden data protection across 3 internal systems.
- Developed keylogger simulation for security research, identifying 6 critical input vulnerability patterns and deploying 4 defensive countermeasures that reduced system attack surface by 35% across 3 internal environments.
- Developed 3 internal security tools and training materials, improving cybersecurity practices for a 15-person engineering team and reducing simulated incident response time by 30%.

Machine Learning Intern, Trestle Academy (4th-IR)

08/2024 – 09/2024

- Developed and deployed a CNN-based pneumonia detection model using TensorFlow and Keras, achieving 95% accuracy on thousands of X-ray images matching radiologist-level diagnostic precision.
- Enhanced model performance through data preprocessing, augmentation, and automated hyperparameter tuning, resulting in a 22% accuracy boost, 20% reduction in overfitting, and 18% faster training.
- Deployed the model with FastAPI for real-time inference, reducing diagnosis time by 20% and improving early detection rates by 15%.

Cloud Engineer Intern, AmaliTech Services GmbH

06/2024 – 08/2024
Remote

- Architected and deployed scalable cloud infrastructures using 30+ AWS services (including EC2, RDS, and S3), boosting system performance by 25% and cutting operational costs by 15%.
- Built and optimized a data pipeline that processed 2TB of Netflix data from Amazon S3, leveraging Amazon QuickSight for analytics and improving content recommendation accuracy by 20%.
- Implemented CI/CD automation with GitHub Actions and AWS, enabling 5 deployments per week, reducing deployment time by 75%, achieving 99.9% system uptime and 80% faster page load speeds.

AWARDS & ACHIEVEMENTS

2nd Place, Best use of Artificial Superintelligence Alliance ,
ETHGlobal, ETHOnline 2025
NeriaAi

2025

Honorary Mention #2 , Lens Spring Hackathon 2025
• Believr

2025

Prize Pool Bounty Winner , Zora Coinathon 2025
• DropSynq

2025

1st Place, Advanced Portfolio Management Track & Brahma ConsoleKit Track,
Safe Agentathon 2025
• NectaFi

2025

SELECTED PROJECTS

EmotiSync , *AI voice companion for emotional well-being*

- Built AI voice companion using OpenAI and Hume AI APIs with sub-800ms response latency, achieving 88% mood detection accuracy across 1,000+ test conversations and reducing user journaling time by 60% through automated summaries.
- Stack: Next.js, TypeScript, OpenAI API, Hume AI API, PostgreSQL, Supabase, Drizzle ORM, Upstash, Zustand, Zod, Bun

Believr , *Decentralized social co-investing platform for creators*

- Built decentralized co-investing platform on Lens Protocol, facilitating \$15K+ in creator investments across 25+ onchain campaigns and connecting 300+ fans to gated creator communities earning Honorary Mention #2 at Lens Spring Hackathon 2025.
- Stack: Next.js, TypeScript, Tailwind, Wagmi, Viem, Zod, IPFS, Lens Protocol

Proof9, *On-chain sound rights platform for music creators*

- Built on-chain sound IP platform using Story Protocol and Yakoa API, verifying 500+ tracks with 92% AI originality detection accuracy and processing automated royalty distributions to 50+ music creators through tamper-proof smart contracts.
- Stack: Next.js, TypeScript, Story Protocol, Yakoa API, Hono, Bun, PostgreSQL, Supabase, IPFS (Pinata), TanStack Query, Tomo SDK, wagmi, viem, zod

EDUCATION

BSc ICT Education, *University of Education, Winneba*

01/2023 – 09/2026

- Progressed from Assistant to Head of Media & Publicity, leading all communications, branding, and digital outreach for the association.
- Organised tech events, workshops, and training sessions that equipped students with practical skills in emerging technologies.
- Grew the association's online presence and community engagement through strategic social media campaigns and content creation.
- Selected as a scholar for the British Council's startup innovation programme, gaining exposure to global entrepreneurship frameworks, innovation methodology, and startup strategy.
- Applied learnings to develop and pitch viable tech-driven solutions addressing local challenges.
- Engaged with a diverse cohort of innovators and mentors from across the region.

Winneba, Ghana

Bsc Computer Science, *University of the People*

01/2023 – 09/2026

Major: Software Engineering & Cloud Computing

Pasadena, CA, USA

Certificates

- AWS Cloud Professional 

References

Eric Annan , *CEO/Founder, Aya HQ*
e.annan@ayahq.com, +233244035369

Prof Delali Kwasi Duke, *Assoc Prof of Computing*, University of Education, Winneba
dkdake@uew.edu.gh, +233540504108